

PROTECTION

The brain has several forms of protection. First and foremost is the skull (see p.48). Between the skull and the grey matter of the cerebrum lie three protective membranes that also protect the spinal cord (see p.98). The dura mater lines the inside of the skull, where it is attached to the bones; it is a thick, inelastic layer that provides support and protection. The arachnoid membrane lies beneath the dura, and was named for its resemblance to a spider's web. The pia mater adheres to the convolutions of the cerebrum, and supplies brain tissue with blood vessels.

Protection also comes from the cerebrospinal fluid (see opposite) in the subarachnoid space, between the arachnoid membrane and the pia mater. It absorbs and disperses excessive mechanical forces that might otherwise cause serious injury. Analyses of its chemical constituents and flow pressure have offered vital clues for diagnosing diseases and disorders of the brain and spinal cord, such as meningitis.

MENINGES OF THE BRAIN

A section through the skull and brain reveals the three meninges. The dura mater lines the skull, and sends four projections inward to give support. The arachnoid membrane cushions the brain, and the pia mater closely envelops the cerebrum.

